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Peer-to-Peer Energy Trading in a Community Microgrid

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***Abstract*—Peer-to-Peer (P2P) energy trading describes flexible energy trades between peers, where the excess energy from many small-scale Distributed Energy Resources (DERs) is traded among local customers. The feasibility of applying P2P energy trading to reduce costs for energy consumers, and to increase income for DER producers in a community microgrid was investigated. Three representative market paradigms were proposed, i.e. bill sharing, mid-market rate and an auction based pricing strategy. Each of them specified detailed business models, local energy exchange prices, as well as quantified individual customer's energy costs. An example of each methodology applied to a residential community microgrid with PV systems, validated the effectiveness of the proposed P2P trading mechanisms and identified the benefits.**

***Index Terms*--Community microgrid, local market paradigm, peer-to-peer energy trading, PV system**

I. Introduction

In recent years Feed-in Tariff (FIT) schemes have been introduced across the world to promote the uptake of a range of small-scale renewable and low-carbon electricity generation technologies. Since the introduction of FIT schemes, a continuous reduction of the tariff rate has been seen, due to falling installation costs and the number of applications for the FIT scheme exceeding government forecasts and funding allocations [1], [2]. With reduced FIT support, the prices for PV owners to sell electricity are becoming lower than retail prices, resulting in a significant reduction in new PV installations. Solutions are required for distributed energy resources (DERs) in order to help governments to fulfill their ambition of reducing Greenhouse Gas emissions.

Using PV electricity on-site at a household level is more attractive than feeding into the grid, due to the difference between electricity selling and buying prices. The self-consumption percentage is defined as the ratio between the energy directly consumed by the PV system owner and the total PV production [3]. Self-consumption depends on daily consumption patterns, and this tends to vary among individual customers because of differences in both electric equipment and habits. In general, for a residential net-zero energy building, the self-consumption is around 20-30% [3]. Load shifting and battery storage were used to improve self-consumption in [4]. It was found that it was possible to increase the self-consumption by 2-15% by demand side management, and between 13 and 24% with a battery storage capacity of 0.5-1 kWh per installed kWp PV power [4].

An alternative solution is to encourage the use of excess energy within a neighborhood, and this can be achieved by Peer-to-Peer (P2P) energy trading. P2P energy trading in a community microgrid describes flexible energy trades between peers, where the excess energy from many small-scale DERs is traded among local customers [5].

The concept of P2P energy trading has been proposed in previous work [5]-[7]. However, assessments are still lacking on the benefit of P2P energy trading being organized as a single entity. Firstly, feasible market paradigms should be devised to carry out a P2P energy trading. Secondly, to the extent that P2P energy trading is able to benefit customers should be investigated. Finally, for network planning purposes, these benefits should be quantified with varying microgrid size (i.e. number of customers), and generation penetration levels. This paper proposed three different P2P market paradigms and applied to an example community microgrid. Their suitability for P2P energy trading in a community energy system, and the potential benefit were assessed.

II. P2P Energy Trading in a Community Grid

Fig. 1(a) presents the trading architecture of a microgrid with *N* households in a conventional electricity market. The solid arrows represent the power flows and the dashed arrows represent the cash flows. Due to the presence of DERs, bi-directional power flows exist. At the time when there is insufficient, or no, PV generation, households buy electricity from suppliers, and when there is excess generation, the surplus is sold to the same supplier.

P2P energy trading provides local and regional energy providers with options to trade energy fairly within the neighborhood, within the community and within the vicinity in the distribution system. This will fundamentally change the current peer to grid paradigm where any surplus of local produce can only be sold to the grid, and transform consumers'

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position from energy and price takers to energy and price makers.

Fig. 1(b) shows the trading architecture of a P2P energy trading in a community microgrid. Here, households are able to exchange energy locally through local buying and selling prices. Therefore P2P energy trading allows local funds to remain within the local economy. Community energy management is required to maintain local energy balancing, where the supply to cover residual demand of the community is purchased at retail prices, and the excess energy is sold to the wholesale market at wholesale prices. Mechanisms should be developed to define the business models, to determine these local energy prices, as well as to quantify energy costs and share benefits [8].

III. P2P MARKET PARADIGMS

Three different representative market paradigms were proposed for a P2P energy trading.

A. *Method* 1*: Bill Sharing* (BS)

The bill sharing method is a cost sharing method that each household pays for their individual electricity use. However, this payment is in the form of a cost share of a single electricity bill of the overall community microgrid, which is recorded by the utility meter at the connection point between the community microgrid and the main grid. This cost is shared according to individual customer's total energy consumption and export. Every customer pays at the same price for each kWh of energy consumption and receives payment at another price for each kWh of export.

The energy use is calculated for individual customers. The electricity demand of household *n* at time *t* is denoted by $L^n(\mathrm{t})$, and the on-site PV generation is denoted by $\mathrm{P}_{PV}^n(\mathrm{t})$. The instantaneously overlapping of load and generation is denoted by $M^n(t)$, which represents the amount of PV generation consumed by the household *n* itself, as shown in

$$M^n(t) = \min\{L^n(\mathrm{t}),\ \mathrm{P}_{PV}^n(\mathrm{t})\} \tag{1}$$

The residual demand of the household *n*, i.e. the amount of electricity to buy (import) is denoted by $P_{im}^n(t)$. The surplus PV generation, i.e. the amount of generation to sell (export) is denoted by $P_{ex}^n(t)$. $P_{im}^n(t)$ and $P_{ex}^n(t)$ were calculated by

$$P_{im}^n(t) = L^n(t) - M^n(t) \tag{2}$$

$$P_{ex}^n(t) = \mathrm{P}_{PV}^n(t) - M^n(t) \tag{3}$$

The total energy that all customers in the community buy from the main grid is denoted by $E_{im,\,total}$, and the total energy the community sell to the main grid is denoted by $E_{ex,\,total}$. They were calculated by

$$E_{im,\,total} = \sum_{n=1}^{N}\left(\sum_{t=0}^{T} P_{im}^n(t) * t\right) \tag{4}$$

$$E_{ex,\,total} = \sum_{n=1}^{N}\left(\sum_{t=0}^{T} P_{ex}^n(t) * t\right) \tag{5}$$

where *N* is the total number of households in the community, and *T* is the time horizon to be considered.

Assume that the price of electricity bought from the grid is

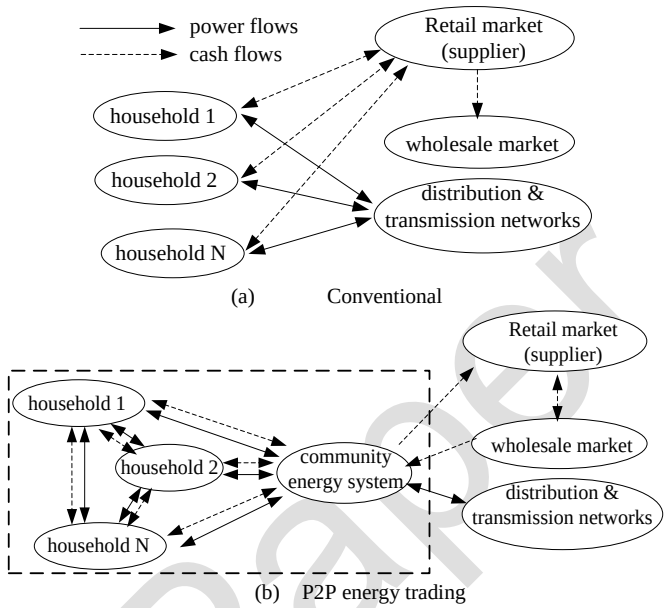

Figure 1. Trading architecture of a community microgrid

c_{BFG}, and the selling price of that sold to the grid is c_{STG}. The electricity bill for each household *n* is

$$C_n = \left(\sum_{t=0}^{T} P_{im}^n(t) * t\right) * c_{BFG} - \left(\sum_{t=0}^{T} P_{ex}^n(t) * t\right) * c_{STG} \tag{6}$$

With P2P energy trading, the energy use of the entire community microgrid is calculated. The instantaneously overlapping of the total generation and total load of the community is denoted by $M^n(t)'$, which represents the amount of PV generation consumed by the community microgrid, as shown in

$$M^n(t)' = \min\{\sum_{n=1}^{N} L^n(\mathrm{t}),\ \sum_{n=1}^{N} \mathrm{P}_{PV}^n(\mathrm{t})\} \tag{7}$$

The residual demand of the community is denoted by $P_{BFG}(t)$. The surplus PV generation is denoted by $P_{STG}(t)$. They are shown in

$$P_{BFG}(t) = \sum_{n=1}^{N} L^n(\mathrm{t}) - M^n(t)' \tag{8}$$

$$P_{STG}(t) = \sum_{n=1}^{N} \mathrm{P}_{PV}^n(\mathrm{t}) - M^n(t)' \tag{9}$$

The total energy a community buys from the main grid is denoted by E_{BFG}, and the total energy a community sells to the main grid is denoted by E_{STG}. They were calculated by

$$E_{BFG} = \sum_{t=0}^{T} P_{BFG}(t) * t \tag{10}$$

$$E_{STG} = \sum_{t=0}^{T} P_{STG}(t) * t \tag{11}$$

Due to the diversity of electricity demands, some excess energy is consumed by their neighbors. Therefore, E_{BFG} is smaller than $E_{im,\,total}$, and E_{STG} is smaller than $E_{ex,\,total}$. Thereafter the P2P electricity buying (or selling) cost to a community is represented by c_{im} (c_{ex}). They are calculated by

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$$c_{im} = c_{BFG} * \frac{E_{BFG}}{E_{im,\,total}} \tag{12}$$

$$c_{ex} = c_{STG} * \frac{E_{STG}}{E_{ex,\,total}} \tag{13}$$

When using the bill sharing method, the electricity bill of individual households was calculated by

$$C_n{}' = \left(\sum_{t=0}^{T} P_{im}^{n}(t) * t\right) * c_{im} - \left(\sum_{t=0}^{T} P_{ex}^{n}(t) * t\right) * c_{ex} \tag{14}$$

B. *Method 2: Mid-Market Rate* (MMR)

The energy exchange price within neighborhood is taken as a value mid-way between the electricity buying price (c_{BFG}) and selling price (c_{STG}). The MMR method assumes that the exchange price is the middle of these two prices, and this is

$$c_{p2p} = \frac{c_{BFG} + c_{STG}}{2} \tag{15}$$

In a community, the local demand and generation vary at different times of a day. When demand is equal to generation, the generation is sold at price of c_{p2p}. At a time when the local generation is higher than demand, the surplus has to be sold to the main grid. When the local generation is lower than demand, some demand is met by local generation, and some is supplied by the main grid. This means the actual buying and selling prices depend on the availability of generation and demand. The actual buying price at time instant *t* is denoted by $c''_{im}(t)$, and the actual selling price is denoted by $c''_{ex}(t)$. Three different scenarios were shown as follows.

1) *PV Generation equal to Demand*

When the PV generation is equal to the demand, the net load of the community is equal to zero. The energy exchange price within a community is c_{p2p}, therefore,

$$c''_{im}(t) = c_{p2p} \tag{16a}$$

$$c''_{ex}(t) = c_{p2p} \tag{17a}$$

2) *PV Generation higher than Demand*

Fig. 2 shows a scenario that the PV generation is higher than the demand. The dark blue line shows the PV generation, and the dashed red line shows the required demand. Therefore, the actual buying and selling prices are

$$c''_{im}(t) = c_{p2p} \tag{16b}$$

$$c''_{ex}(t) = \frac{\sum_{n=1}^{N} L^{n}(\mathrm{t}) * c_{p2p} + (\sum_{n=1}^{N} \mathrm{P}_{PV}^{n}(\mathrm{t}) - \sum_{n=1}^{N} L^{n}(\mathrm{t})) * c_{STG}}{\sum_{n=1}^{N} \mathrm{P}_{PV}^{n}(\mathrm{t})} \tag{17b}$$

3) *PV Generation lower than Demand*

Fig. 3 shows a situation that the PV generation is lower than the demand. The dark blue line shows the PV generation, and the dashed red line shows the required demand. Therefore, the actual buying and selling prices were represented by

$$c''_{im}(t) = \frac{\sum_{n=1}^{N} \mathrm{P}_{PV}^{n}(\mathrm{t}) * c_{p2p} + (\sum_{n=1}^{N} L^{n}(\mathrm{t}) - \sum_{n=1}^{N} \mathrm{P}_{PV}^{n}(\mathrm{t})) * c_{BFG}}{\sum_{n=1}^{N} L^{n}(\mathrm{t})} \tag{16c}$$

$$c''_{ex}(t) = c_{p2p} \tag{17c}$$

When using a mid-market rate method, the electricity bill of individual households was calculated by

$$C_n{}'' = \sum_{t=0}^{T} (P_{im}^{n}(t) * t * c''_{im}(\mathrm{t})) - \sum_{t=0}^{T} (P_{ex}^{n}(t) * t * c''_{ex}(\mathrm{t})) \tag{18}$$

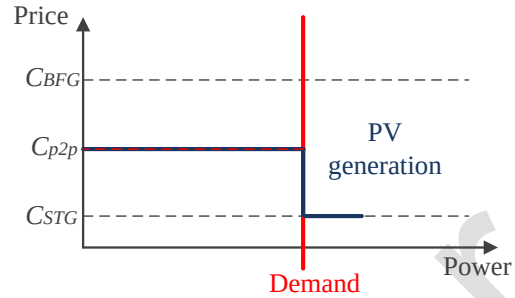

Figure 2. P2P energy trading with mid-market rate: PV generation>demand

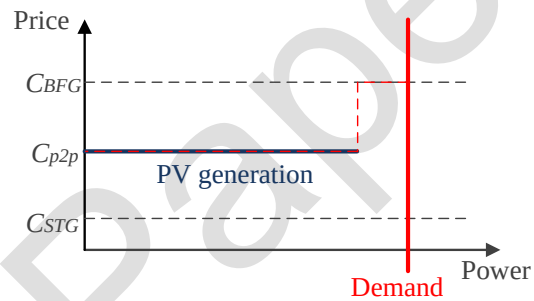

Figure 3. P2P energy trading with mid-market rate: PV generation<demand

C. *Method 3: Auction-based Pricing Strategy* (APS)

APS is a method that utilizes an auction market for local demand and generation. Each household plays an active role in providing bids or offers of their demand or generation. These offers and bids are put together and managed using pre-defined rules to allocate and define the price strategy.

The process of P2P energy trading using APS is shown in Fig. 4. At time t, each household announces an estimated clearing price based on the bidding/offering and clearing price at the previous time (i.e. *t*-1) using a recursive least square (RLS) method. Each household's role in terms of buying or selling is determined by their net load. If PV generation is smaller than demand, the household is a buyer. The household provides bids with the amount of electricity it has to buy, and the household's estimated clearing price is the bidding price. If PV generation is higher than demand, the household is a seller. The household provides offers with the amount of electricity it has to sell, and the household's estimated clearing price is the offer price. Then these bids/offers are put in ascending order. The clearing price is determined by the actual demand and generation in the community. Next, the absolute value of the difference between buyers biding prices and the clearing price is calculated. This value determines the corresponding customer's priority to obtain the cheaper offers. The smaller the absolute value, the higher the priority. Finally the actual buying and selling prices for time *t* is determined. This process then runs for the next time instant, *t*+1.

A recursive least square (RLS) method was used for the households to estimate the clearing price, and this is

$$\hat{c}_g^n(t+1) = \hat{c}_g^n(t) + \gamma^n(c_g^n(t) - \hat{c}_g^n(t)) \tag{19}$$

where $\hat{c}_g^n(t)$ is the estimate of the clearing price at time *t* for the household *n*. $c_g^n(t)$ is the clearing price of the market at the

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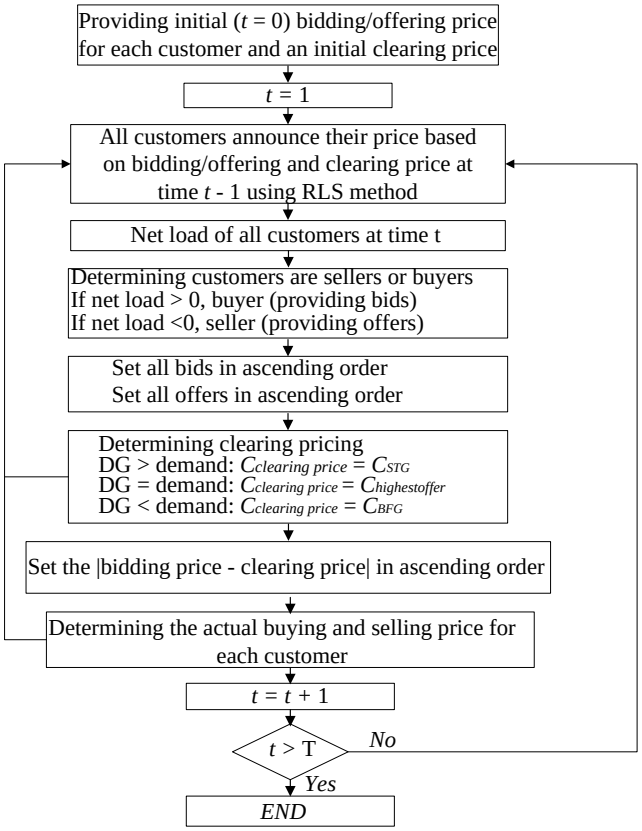

Figure 4. Process of P2P energy trading - auction based pricing strategy

time t. γ^n is the learning (filtering) rate. This learning mechanism is based on a 1-D recursive least squares [9].

Clearing price is determined by the availability of demand and generation. Three scenarios were presented as follows.

1) PV Generation equal to Demand

Fig. 5 shows a scenario that the PV generation is equal to the demand. The clearing price is equal to the highest offer among all offers provided by sellers.

2) PV Generation higher than Demand

Fig. 6 shows a scenario that the PV generation is higher than the demand. The surplus is sold to the main grid at the selling price, so the clearing price is equal to $C_{STG}(t)$.

3) PV Generation lower than Demand

Fig. 7 shows a scenario that the PV generation is lower than the demand. The residual demand is supplied by the main grid at the buying price, so the clearing price is $C_{BFG}(t)$.

IV. CASE STUDY

Fig. 8 shows a community microgrid with 10 households, 5 of which have PV systems installed. The community grid is connected to the main grid through a utility meter. Daily demand and PV generation profiles with 1-min resolution were obtained using Centre for Renewable Energy Systems Technology (CREST) tool [10]. The load of each individual household is modelled realistically considering occupancy, the associated use of electrical appliances with randomness of human activities. The PV systems are considered to have the same generation profile, due to the relatively small area of the

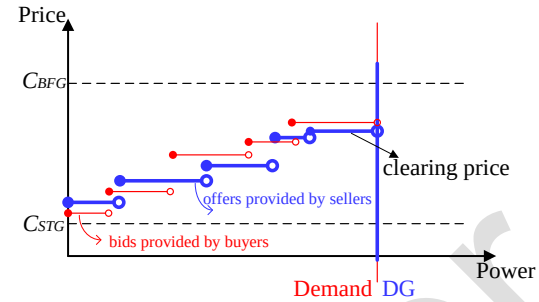

Figure 5. Auction based pricing strategy: PV generation = demand

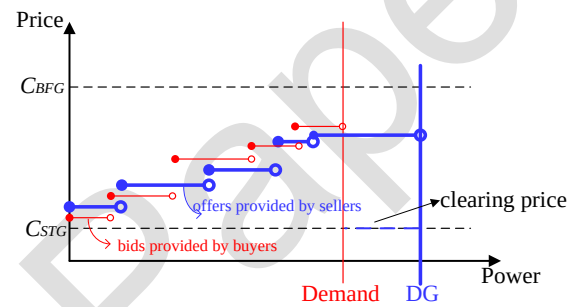

Figure 6. Auction based pricing strategy: PV generation > demand

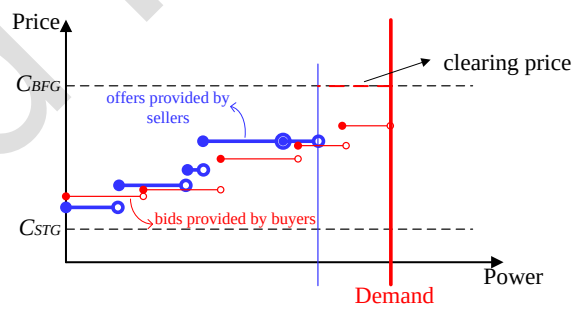

Figure 7. Auction based pricing strategy: PV generation < demand

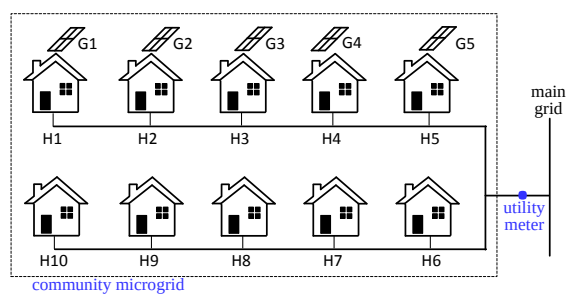

Figure 8. An example community microgrid

microgrid. The generation peak is randomly selected from a range of 2.0 to 3.5 kWp.

Table I presents electricity price and bill for each individual households on a week day. The grid buying price was 15 pence per kWh and the selling price was 5 pence per kWh. Within a conventional market, the total bill for all households was £6.99 per day, and this was reduced to £5.04 when using P2P energy trading regardless of the market paradigm used.

The total daily energy bill was the same for the three market paradigms, because that the demand response was not

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TABLE I
ELECTRICITY PRICE AND BILL FOR EACH INDIVIDUAL HOUSEHOLDS ON A WEEK DAY

	Energy		Base: conventional			Method 1: BS			Method 2: MMR			Method 3: APS		
	Buy	Sell	Buy	Sell	Cost(£)	Buy	Sell	Cost(£)	Buy	Sell	Cost(£)	Buy	Sell	Cost(£)
	kWh		Price (£/kWh)			Price (£/kWh)			Price (£/kWh)			Price (£/kWh)		
Customer 1	3.77	-9.55	0.15	0.05	**0.09**	0.105	0.032	**0.09**	0.139	0.067	**-0.12**	0.144	0.061	**-0.05**
Customer 2	11.3	-9.07	0.15	0.05	**1.24**	0.105	0.032	**0.89**	0.137	0.070	**0.91**	0.137	0.065	**0.95**
Customer 3	5.94	-8.46	0.15	0.05	**0.47**	0.105	0.032	**0.35**	0.135	0.068	**0.22**	0.129	0.063	**0.23**
Customer 4	4.90	-13.8	0.15	0.05	**0.04**	0.105	0.032	**0.08**	0.140	0.068	**-0.25**	0.145	0.062	**-0.15**
Customer 5	5.47	-11.9	0.15	0.05	**0.22**	0.105	0.032	**0.19**	0.144	0.069	**-0.03**	0.141	0.064	**0.01**
Customer 6	5.51	0	0.15	0.05	**0.83**	0.105	-	**0.58**	0.124	-	**0.69**	0.106	-	**0.59**
Customer 7	7.63	0	0.15	0.05	**1.14**	0.105	-	**0.80**	0.135	-	**1.03**	0.133	-	**1.02**
Customer 8	6.78	0	0.15	0.05	**1.02**	0.105	-	**0.71**	0.132	-	**0.89**	0.122	-	**0.82**
Customer 9	4.50	0	0.15	0.05	**0.67**	0.105	-	**0.47**	0.132	-	**0.59**	0.126	-	**0.57**
Customer 10	8.42	0	0.15	0.05	**1.26**	0.105	-	**0.88**	0.132	-	**1.11**	0.124	-	**1.04**
Total	64.2	-52.8	0.15	0.05	**6.99**	0.105	0.032	**5.04**	0.135	0.068	**5.04**	0.130	0.063	**5.04**

BS: bill sharing; MMR: mid-market rate; APS: auction-based pricing strategy; Negative cost means income.

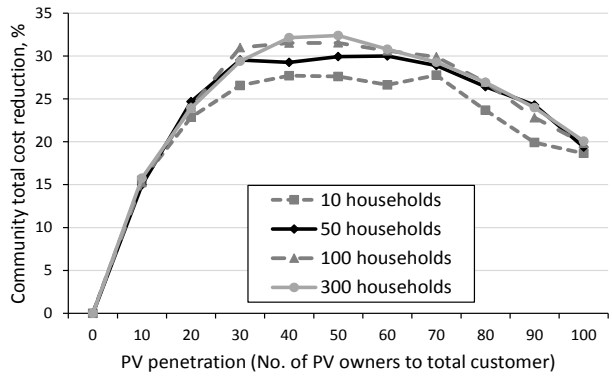

Figure 9. Commnity energy cost reduction vs PV generation

considered. For the bill sharing method, electricity is paid without considering the time of use. This is not likely to change customers' electric use pattern. For the MMR method, buying and selling prices are varied over time depending on the local availability. Consumption coinciding with local generation will pay less, and vice versa. This will stimulate flexible demand to respond to local generation, and flexible DER (such as small hydro, battery storage) to generate more during peak load time. For the auction based method, the more accurate estimate a customer is able to make, the better tariff the customer has. This will allow customers not only to use their flexible demand or DER to respond to price, but also predict their community energy resources to manage their own energy use. Therefore, there will be more cost reduction using the MMR and auction-based method than the bill sharing method.

Fig. 9 presents the energy cost reduction of a community with different household numbers (from 10 to 300 households) and different PV penetrations (0 to 100%). It is shown that from 0 to 50% PV penetration, the percentage of energy cost reduction continues to increase. From 60 to 100% PV penetration, this cost reduction decreases. Due to the diversity of electricity demand, the more customers in a community microgrid, the larger percentage of energy cost reduction.

V. CONCLUSIONS

Three different market paradigms, i.e. bill sharing, mid-market rate and auction based pricing strategy, were proposed to apply P2P energy trading in a community microgrid. For each of these market paradigms, business models were specified, local energy exchange prices were defined, and the feasibility of P2P energy trading to reduce consumer energy costs and to increase income for DER producers was assessed.

It is found that with a moderate level of PV penetration, P2P energy trading resulted in a reduction of community energy costs by ~30%. This cost reduction resulted from the diversity of demands. This work provides an encouragement for P2P energy trading, and further benefits are expected when these P2P trading paradigms stimulate flexible demand response to be added to these pricing schemes. Though this work was applied to a residential community micrgrid with PV technology, these P2P trading mechanisms are applicable to larger scale of community grids with other DER technologies.

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